
AKSHARA REDDY ARRABOLU

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PROFESSIONAL SUMMARY

Master's graduate in Computer Science with hands-on experience in machine learning, cloud computing, and applied AI systems. Proven ability to build data-driven solutions, deploy scalable systems, and translate technical insights into real-world impact. Strong communicator with experience working on end-to-end projects across AI, analytics, and cloud platforms.

RECENT EXPERIENCE

GRADUATE ENGINEER TRAINEE- CLOUD COMPUTING (Internship)

Feb 2023 - Apr 2023

LTIMindtree, India

- Completed LTIMindtree IGNITE training in Cloud Computing with an 87% score, gaining foundational and practical expertise.
- Assisted in designing and deploying scalable cloud infrastructure using AWS, Azure, and Google Cloud.
- Automated resource provisioning with Terraform and AWS CloudFormation, improving deployment efficiency.
- Supported migration of legacy systems to cloud-native environments with minimal downtime.
- Explored containerization tools like Docker and Kubernetes for scalable application deployment.

ACADEMIC PROJECTS

GAIA – Generative AI for Automated Academic Assessment (AWS Cloud)

Designed and deployed a cloud-based Generative AI platform to automate grading, feedback generation, and AI-content detection. Built a full-stack role-based application using FastAPI and React, deployed on AWS serverless services. Reduced grading time by ~70% and achieved a 0.87 correlation with human grading.

Tech Stack: Python, OpenAI API, FastAPI, React, AWS (Lambda, API Gateway, S3, RDS), PostgreSQL

Dynamic Hotel Recommendation System Using NLP

Developed a personalized hotel recommendation system using NLP-based sentiment analysis of user reviews. Implemented a hybrid model combining content-based and collaborative filtering to improve recommendation accuracy and personalization.

Tech Stack: Python, NLP, NLTK, Pandas, NumPy, Matplotlib

Employee Attrition Prediction Using Machine Learning

Built predictive models using Logistic Regression, Decision Trees, and Random Forests to analyze employee attrition patterns from IBM HR datasets. Performed data preprocessing, feature engineering, and model evaluation to identify key drivers such as job satisfaction and work-life balance. Designed interactive dashboards to visualize insights and support data-driven retention strategies.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Tableau

Image Classification Using Convolutional Neural Networks

Designed and trained a Convolutional Neural Network (CNN) for multi-class image classification, achieving 95% accuracy on the test dataset. Applied data augmentation, hyperparameter tuning, and early stopping to improve model generalization and prevent overfitting. Optimized the training pipeline to balance accuracy and computational efficiency.

Tech Stack: Python, TensorFlow, Keras, OpenCV, Matplotlib

SKILLS

Programming: Python, Java, C, Bash, HTML, CSS, JavaScript

Frameworks/Libraries: TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy, Django

Cloud/DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, Linux, Git, CI/CD

AI/Data Science: Machine Learning, Deep Learning, NLP, Data Mining, AI, Data Visualization, Data Analysis (EDA, Statistical Modeling), Excel (Pivot Tables, Charts, Power Query)

Development: Web & Mobile Development, Software Engineering, OOP, Data Structures, Algorithms

Databases: MySQL, SQLite, MongoDB

Soft Skills: Leadership, Problem Solving, Communication, Team Collaboration, Adaptability

EDUCATION

MASTER OF SCIENCE IN COMPUTER SCIENCE

Western Michigan University, Kalamazoo, USA

Aug 2023 - Dec 2025

GPA – 3.6/4

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE ENGINEERING

CMR Institute of Technology, Hyderabad, India

Aug 2019 - May 2023

LEADERSHIP & WORKSHOP

Student Coordinator – National Service Scheme (NSS)

Nov 2019 – May 2023

CMR Institute of Technology, India

- Led and coordinated student volunteer teams for community service initiatives. Organized and executed multiple social impact projects, enhancing community engagement. Facilitated communication between faculty and volunteers, and managed resource allocation. Mentored junior volunteers and implemented effective collaboration strategies.

Workshop:

Fundamentals of Deep Learning – NVIDIA Deep Learning Institute

Feb 2023

Gained foundational knowledge of neural networks and deep learning through hands-on training.